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End Semester Examination 2025

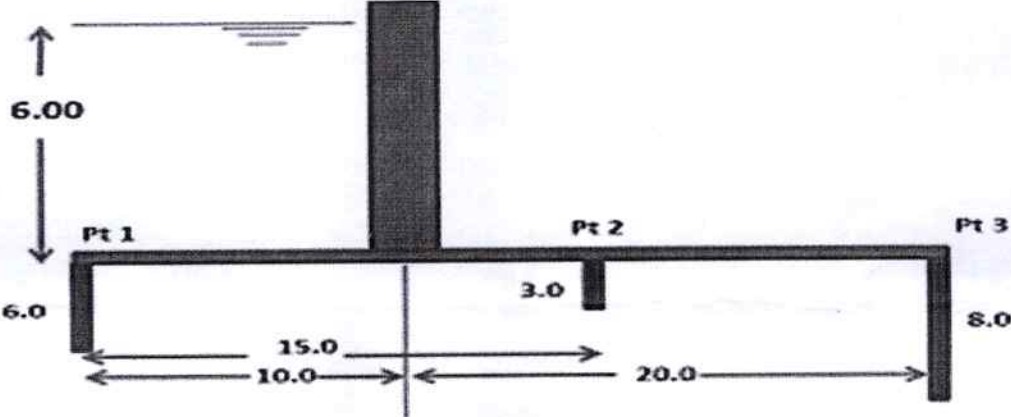
Name of the Course: **B.Tech. Civil Engineering**
 Name of the Paper: **Water Resources Engineering II**
 Time: 3 Hours

Semester: **VI**
 Paper Code: **TCE-603**
 Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each sub question carries 10 marks

Q1	(20marks)	CO 1
(a)	Define elementary profile of a gravity dam. Obtain an expression for the base width of the elementary profile for no tension	
(b)	Explain briefly with neat sketch's the different forces that may act on a gravity dam	
(c)	Determine the wave force and free board given: Wind velocity = 100 Km/hr Fetch = 34 Km Average depth of reservoir along fetch = 18 m	
Q2	(20marks)	CO 2
(a)	Discuss briefly the various causes of failure of earthen dams.	
(b)	Prove that the most economical section of an arch dam has a central angle of $133^{\circ}34'$	
(c)	Calculate the seepage through an earthen dam resting on an impervious foundation. The relevant data are given below Height of the dam = 60.0 m Upstream slope = 2.5 H: 1 V Downstream slope = 2.5 H: 1 V Freeboard = 2.5 m Crest width = 8.0 m Length of drainage blanket = 120.0 m Coefficient of permeability of the embankment material in X-direction = 8×10^{-7} m/s Y-direction = 2×10^{-7} m/s	
Q3	(20marks)	CO 3
(a)	Enumerate the various factors affecting the design of energy dissipaters. Give the classification of energy dissipaters based on the position of tail water rating curve w.r.t. jump height curve.	
(b)	Enumerate the important types of spillway gates. Describe with a neat sketch the construction and working of Drum gate.	

(c)	A spillway is to discharge $20 \text{ m}^3/\text{sec}/\text{m}$ length. If the maximum water level is to be 50 m, above river bed, obtained the profile of the spillway crest, using USWES method. Take the u/s face as vertical & d/s has a slope of 0.8 H : 1.0 V.	
Q4		
(a)	Explain the main causes of failure of hydraulic structures founded on pervious foundations. What remedies would you suggest to prevent them?	CO 4
(b)	Draw a neat sketch of a typical layout of diversion head works and explain the function of its component parts.	
(c)	The figure shows the section of hydraulic structure on permeable foundation. Calculate the average hydraulic gradient by Bligh's creep theory and Lane's creep theory and find uplift pressure and thickness of floor at a distance 10 m. (starting point of floor i.e. Pt 1). Also locate the point where required floor thickness is same for both the theory. Take $s = 2.24$ 	
Q5		
(a)	Derive a relationship for spacing of deep tile drains for drainage of irrigated areas	CO 5
(b)	Discuss the necessity of river training work and enumerate the various works associated with it. Explain any one method with the help of neat sketch.	
(c)	State the advantages and disadvantages of hydropower generation over thermal and nuclear power generation	